AR/VR ASSIGNMENT – 3

The goal of this project was to create an immersive WebXR-based VR application using A-Frame featuring interactive dynamic pallets (cards). The application demonstrates 3D interaction, flip animations, and immersive navigation inside a futuristic virtual gallery environment.

Task:

I created a VR gallery environment using A-Frame where five interactive quote pallets are positioned inside a neon-themed 3D room. Each pallet displays a famous quote on the front side and the corresponding author's name on the back side.

The pallets dynamically flip along either the X-axis or Y-axis when clicked by the user. A custom A-Frame component named flip-on-interaction was used to control the flipping logic and animation handling. Random axis selection was implemented using JavaScript so that different pallets flip in different directions.

To improve interactivity, hover effects were added where the pallets slightly enlarge and glow when the cursor is placed over them. Smooth animations were created using A-Frame's built-in animation system with easing functions for realistic motion.

The project also includes:

- Interactive flip animations
- Hover scaling effects
- Random X/Y axis flipping
- Neon Lighting environment
- Glass morphism pallet design
- WASD movement controls
- Mouse-based viewing controls
- Immersive VR gallery experience

Environment Design

The environment was designed as a futuristic neon art gallery with dark walls, reflective flooring, ambient lighting, and glowing LED strips. Cyan and pink glowing borders were used on the pallets to create a modern cyberpunk visual style. Ambient and spotlight lighting were used to improve visibility and create depth inside the virtual room.

Flip Animation Logic

The flip animation was implemented using a custom A-Frame component called flip-on-interaction. Each pallet maintains two states:

- isFlipped
- isAnimating

This isAnimating state prevents rapid repeated clicks during animation execution. When a click event occurs, the pallet smoothly rotates by 180 degrees either along the X-axis or Y-axis depending on the randomly assigned axis.

Text Orientation Handling

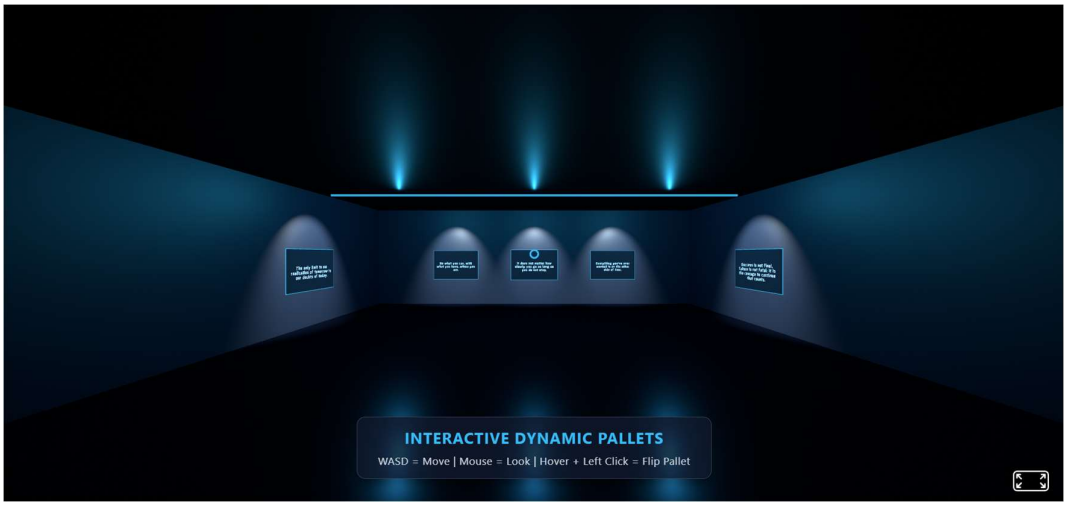
To ensure the back-side text remains readable and not mirrored after flipping, rotation correction was applied dynamically. For Y-axis flips, the back pallet face is rotated by ‘0’ ‘180’ ‘0’, while for X-axis flips, it is rotated by ‘180’ ‘0’ ‘0’. This correction ensures that the author’s name always appears properly oriented after the pallet flips.

Technologies Used:

- HTML
- CSS
- JavaScript
- A-Frame
- WebXR
- A-Frame Environment Component

Screenshots

Interactive VR Gallery View



Flipped Pallet Showing Author Name

Conclusion

This project successfully demonstrates an immersive VR-based interactive quote gallery using A-Frame and WebXR concepts. The pallets dynamically respond to user interaction through hover and click events while maintaining smooth animations and proper text orientation. The futuristic neon environment and interactive design enhance the overall VR experience.